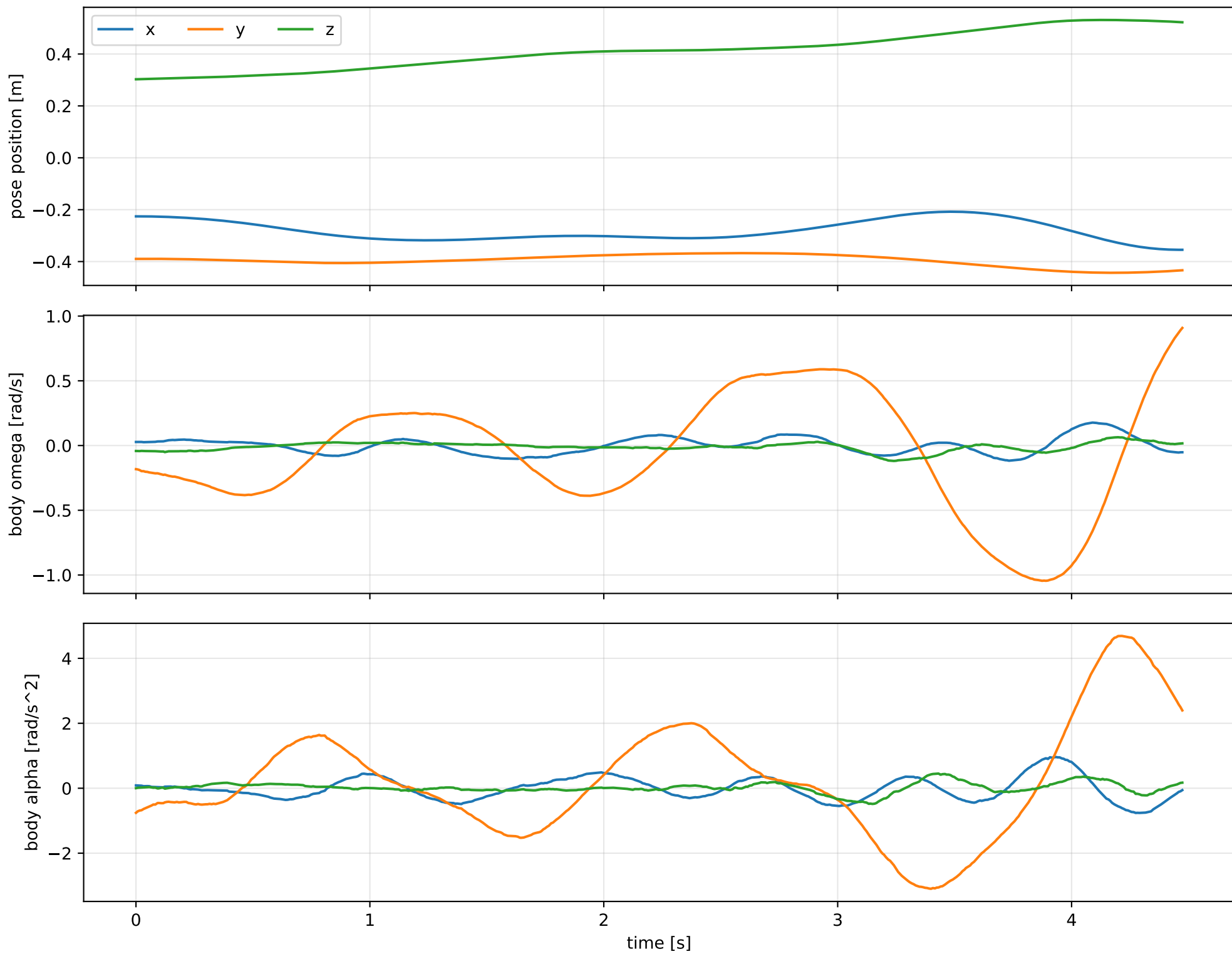
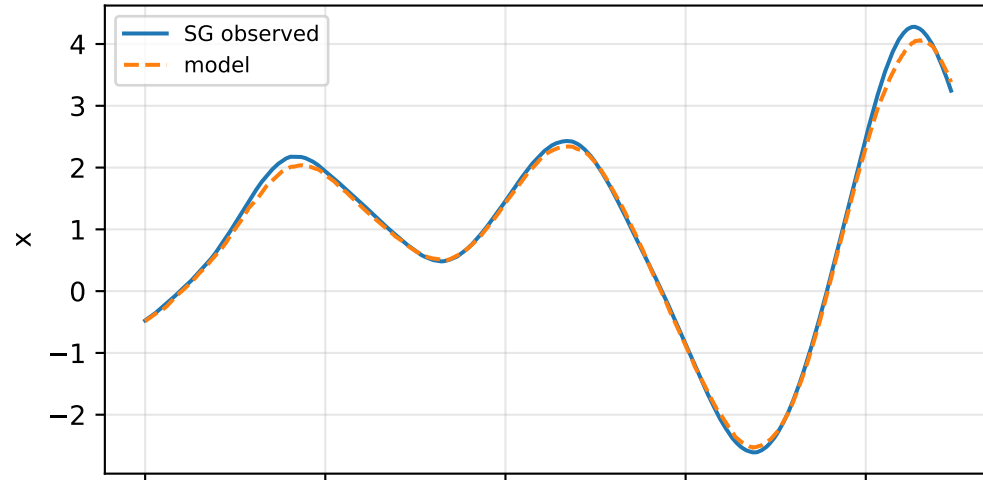


kkt_scale_positive: SG trajectory and derived motion

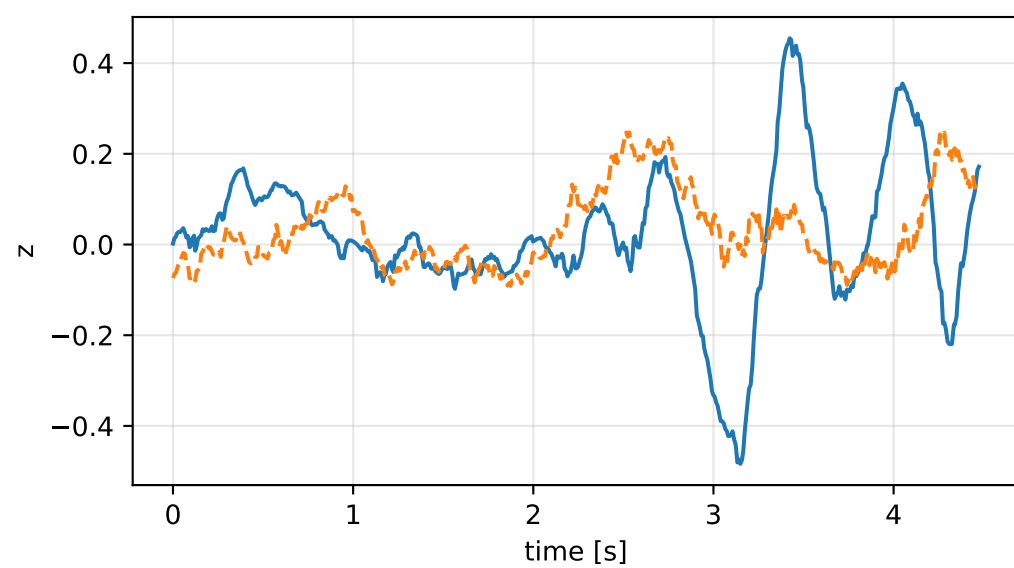
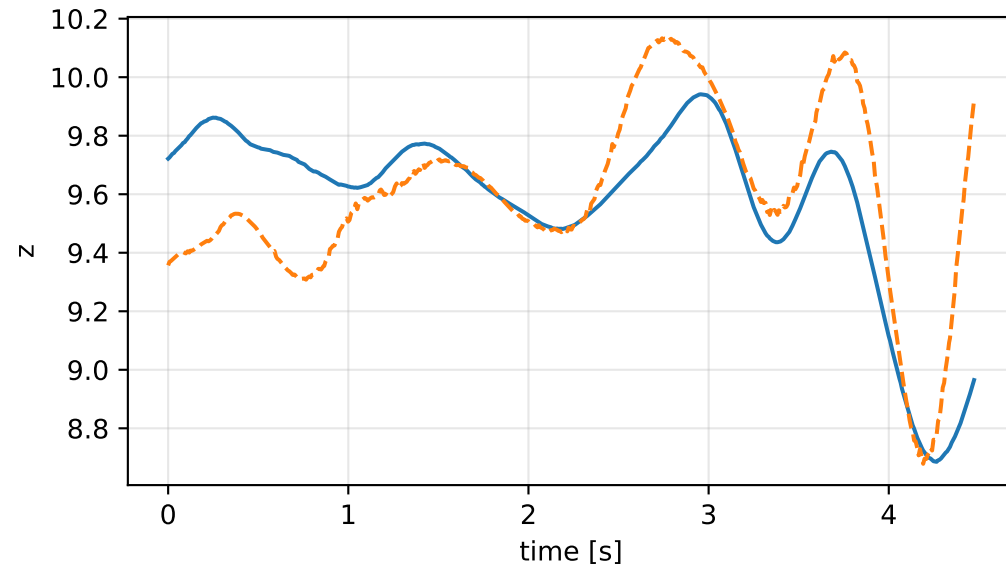
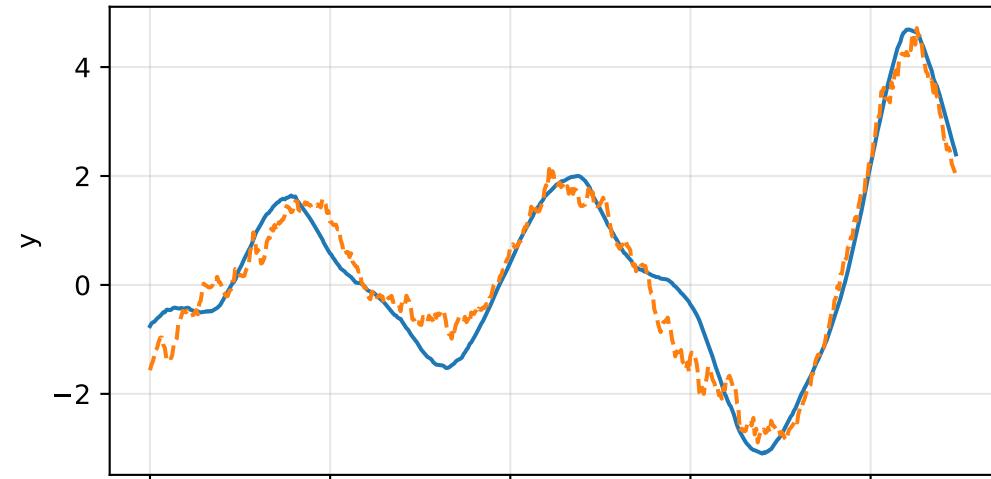
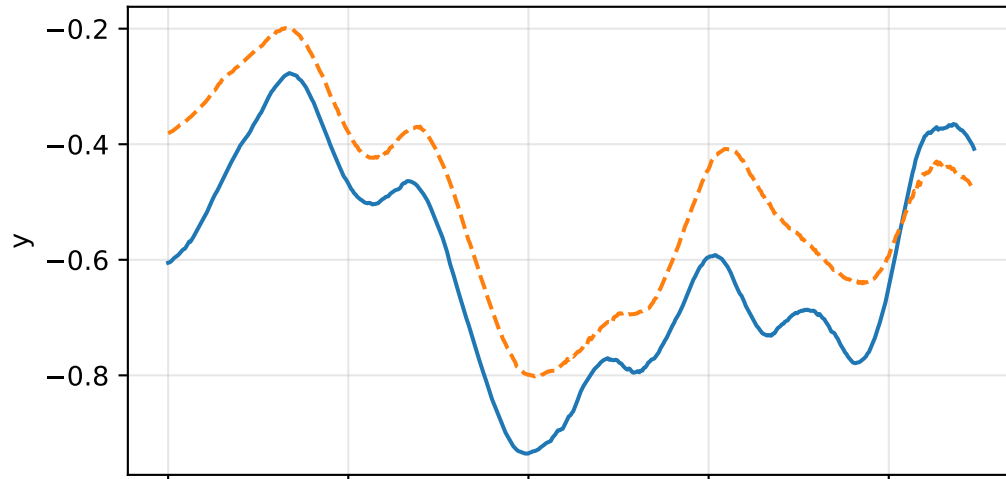
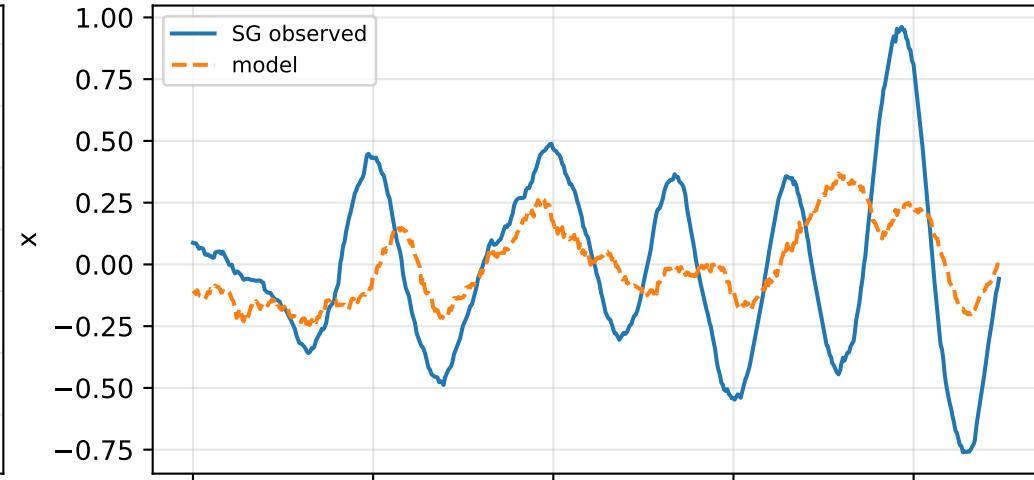


kkt_scale_positive: acceleration residual objective

specific acceleration [m/s^2]

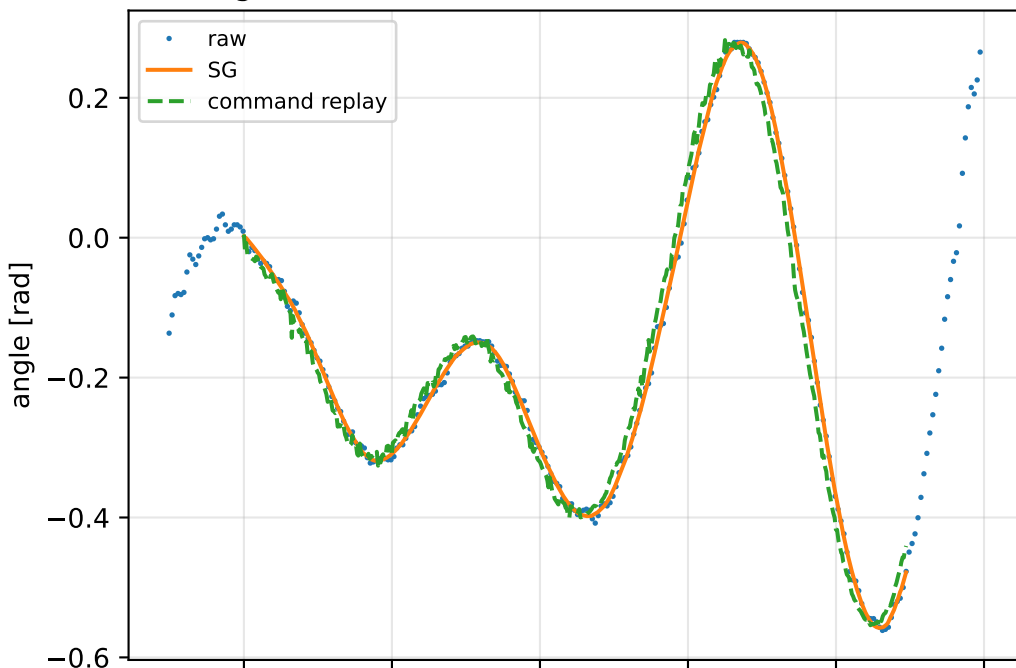


angular acceleration [rad/s^2]

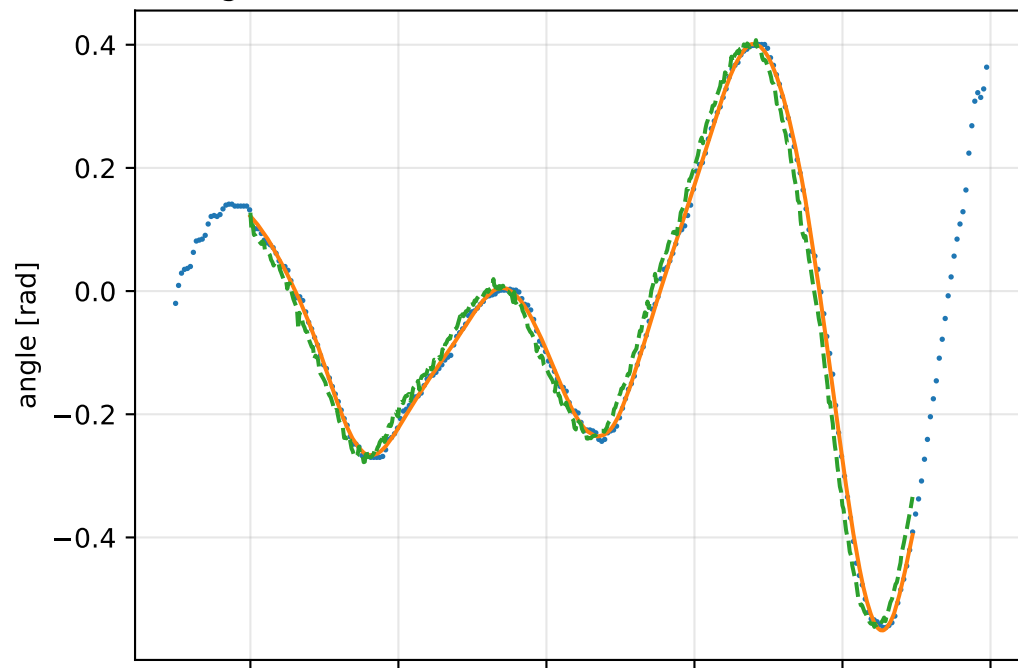


kkt_scale_positive: gimbal measurement audit (objective=measured_sg)

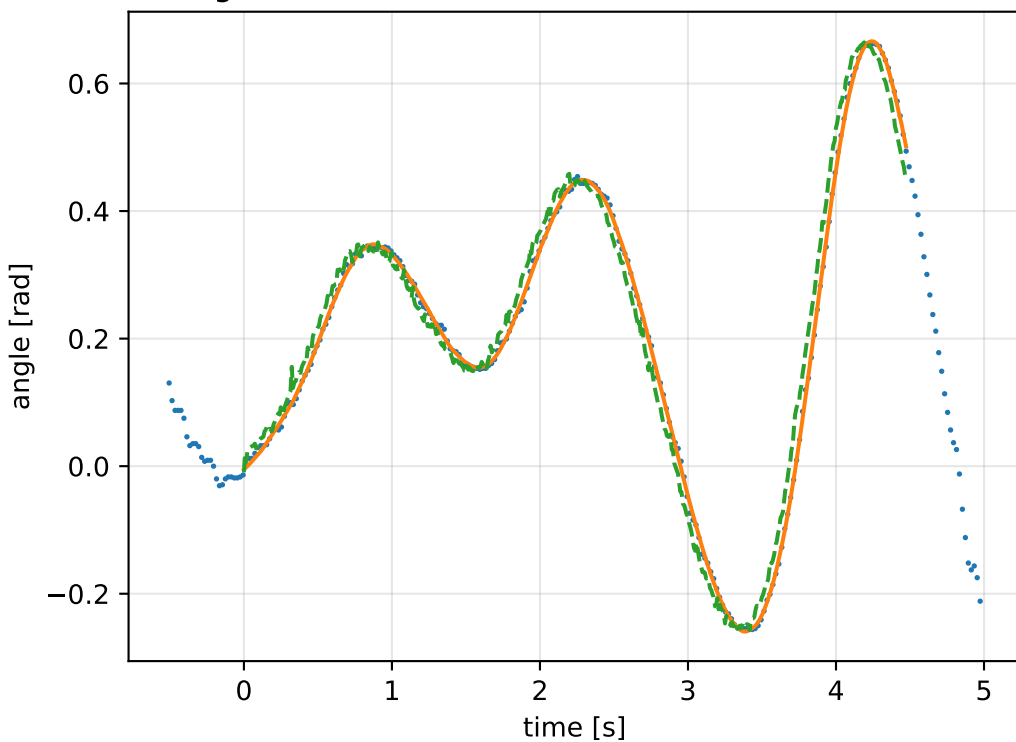
gimbal 1: RMSE=0.027, max=0.07529 rad



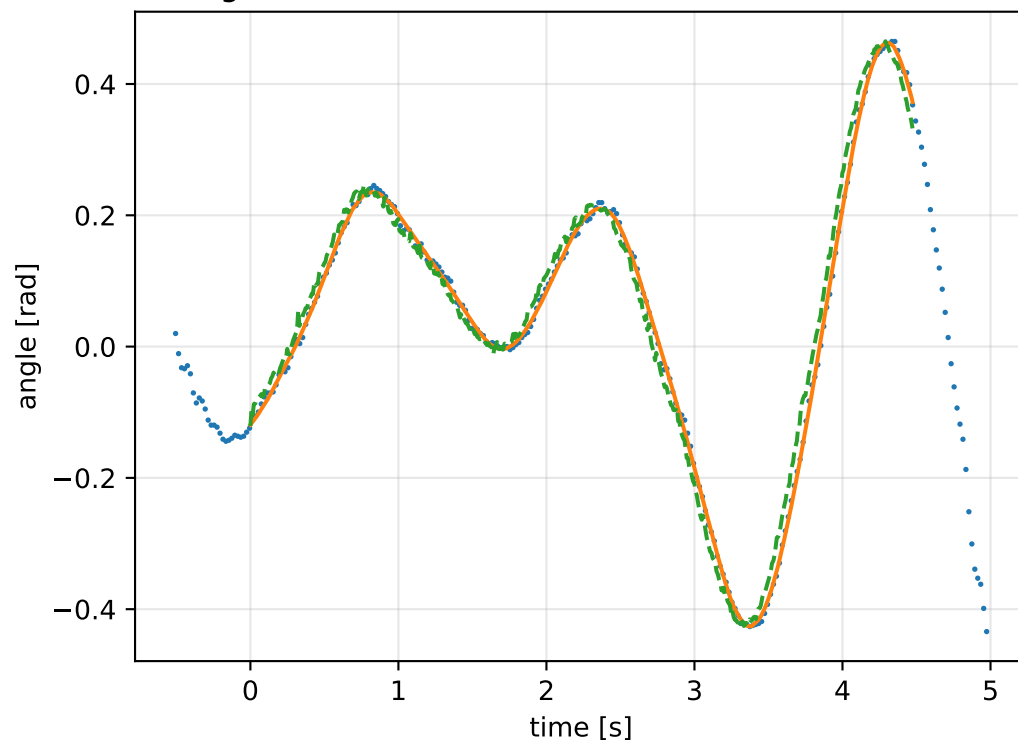
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

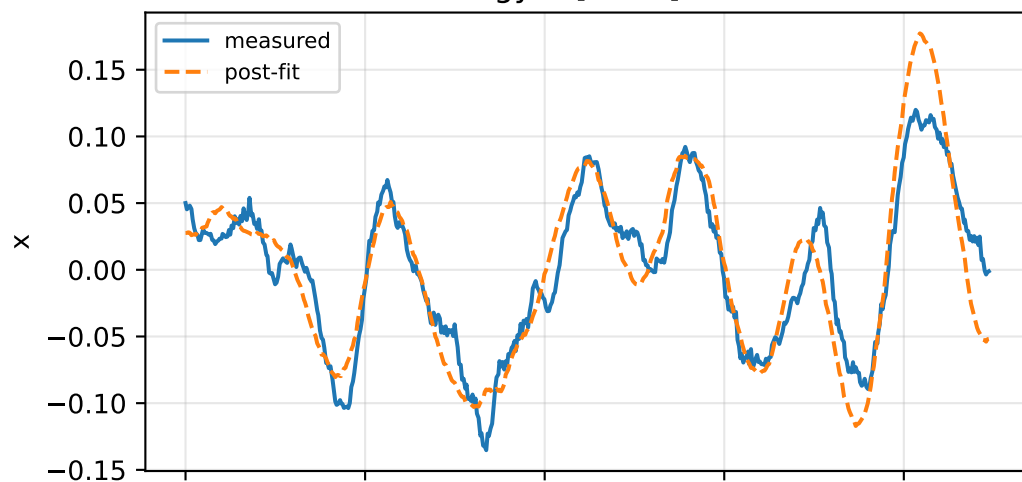


gimbal 4: RMSE=0.02598, max=0.07261 rad

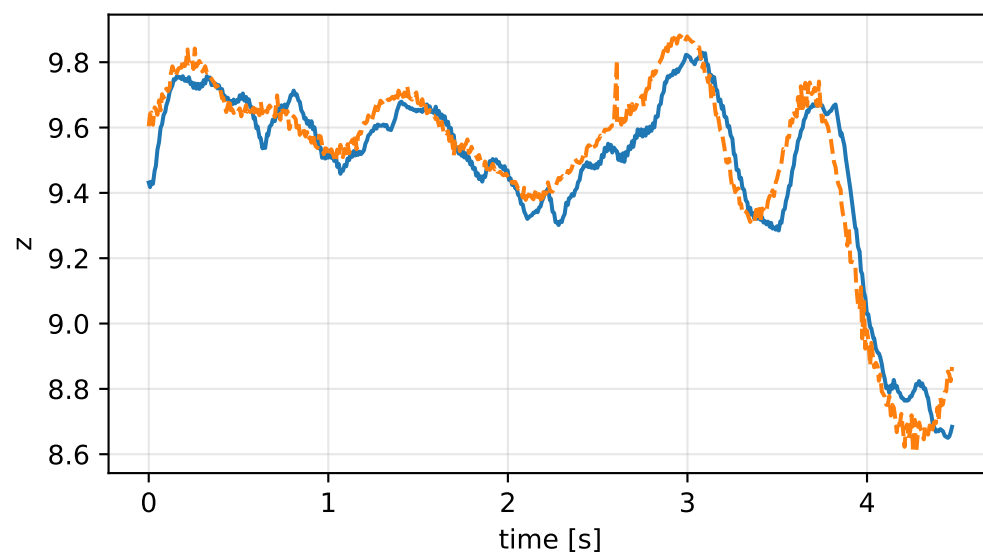
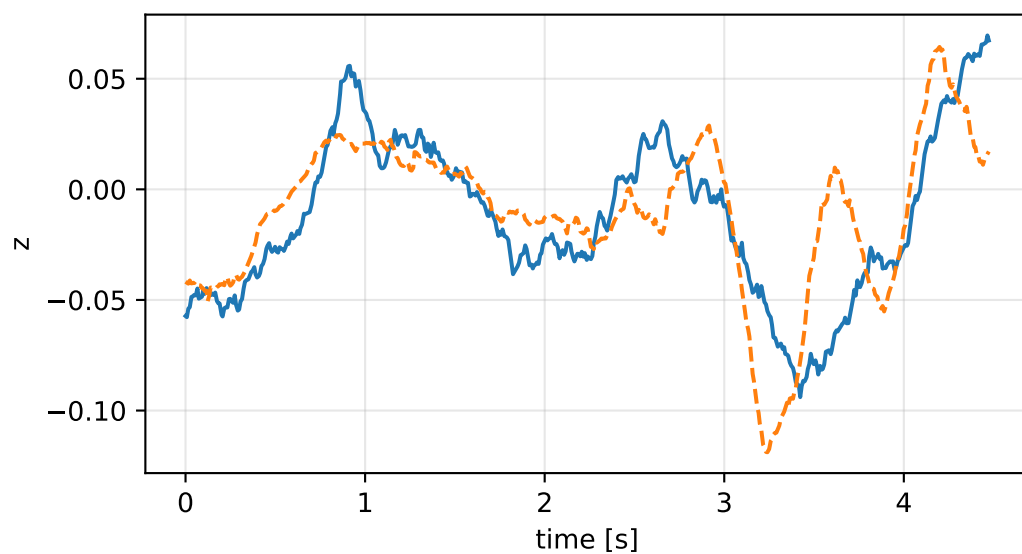
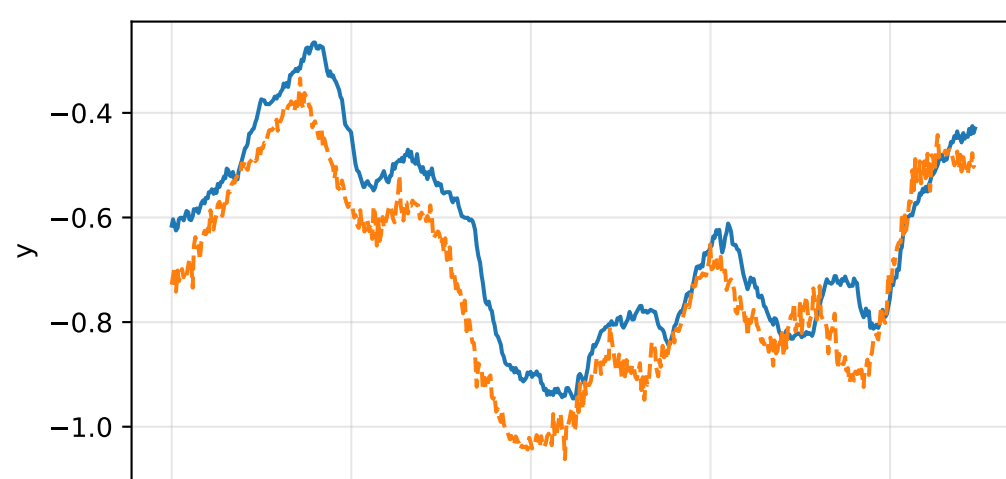
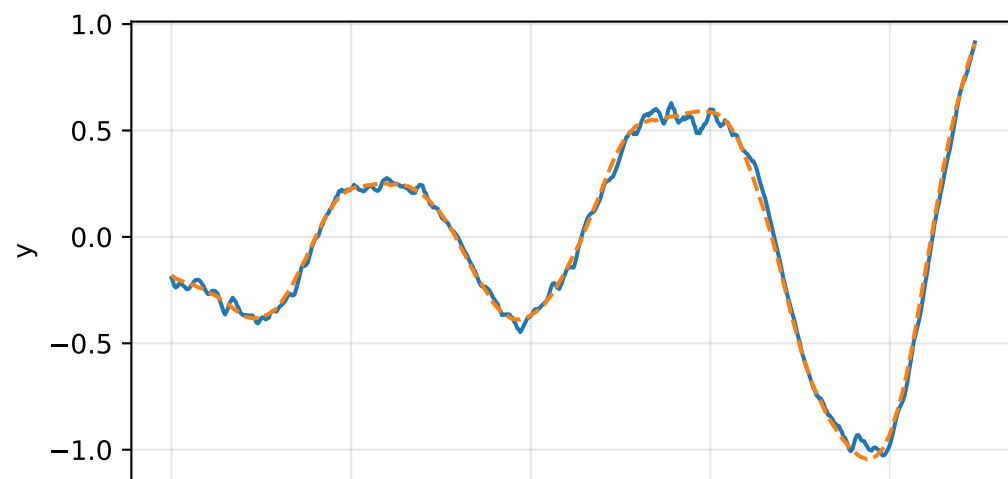
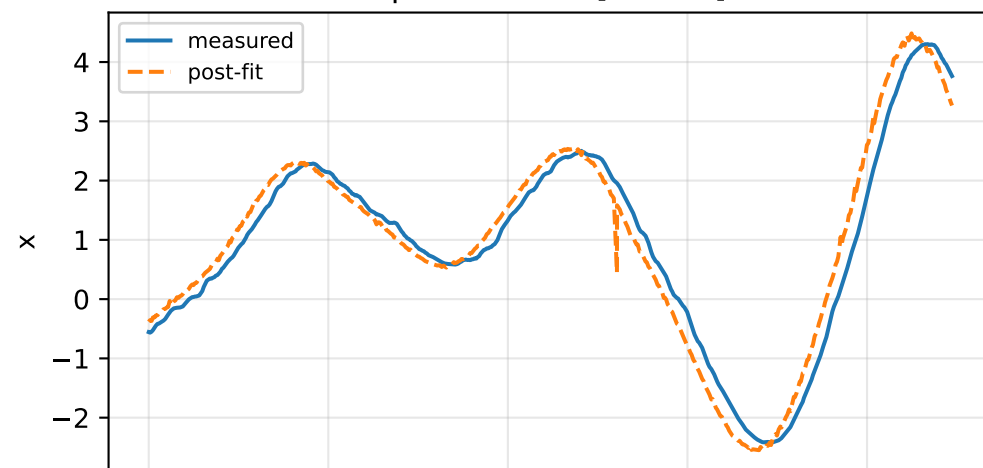


kkt_scale_positive: IMU comparison (diagnostic only)

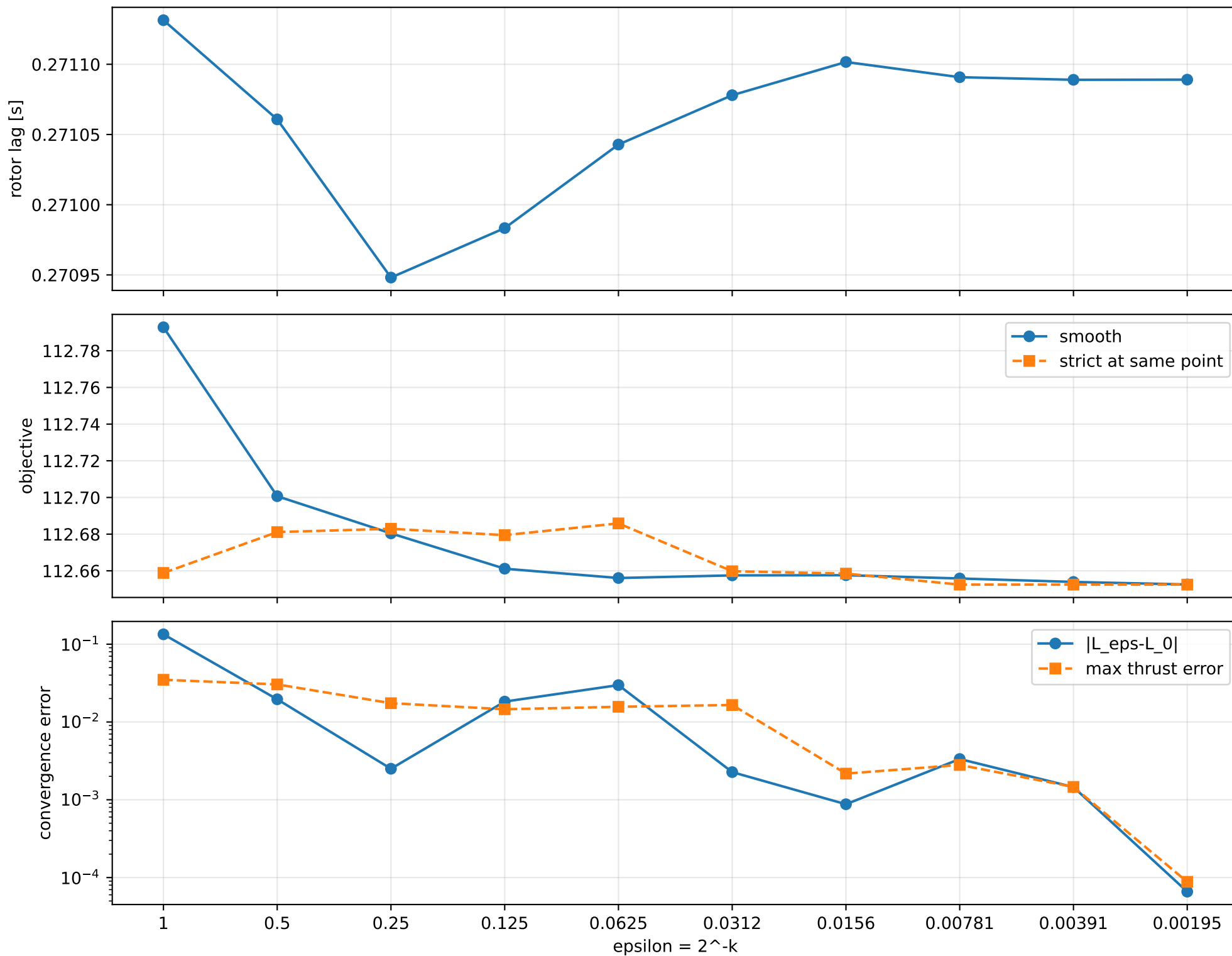
gyro [rad/s]



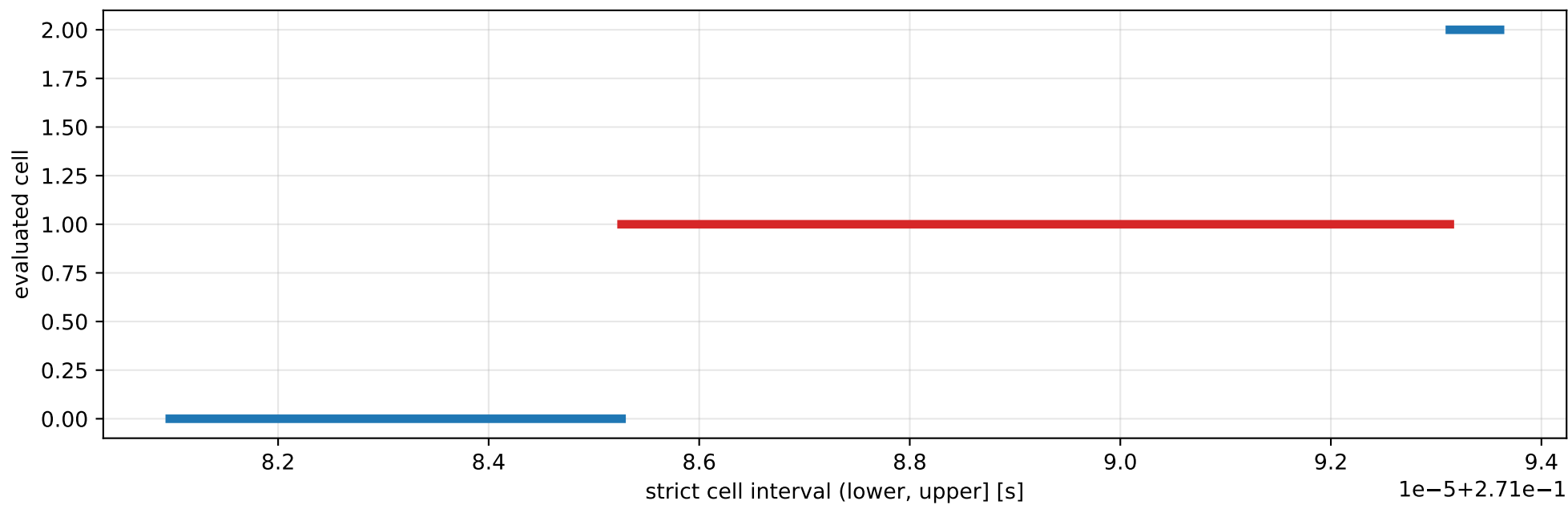
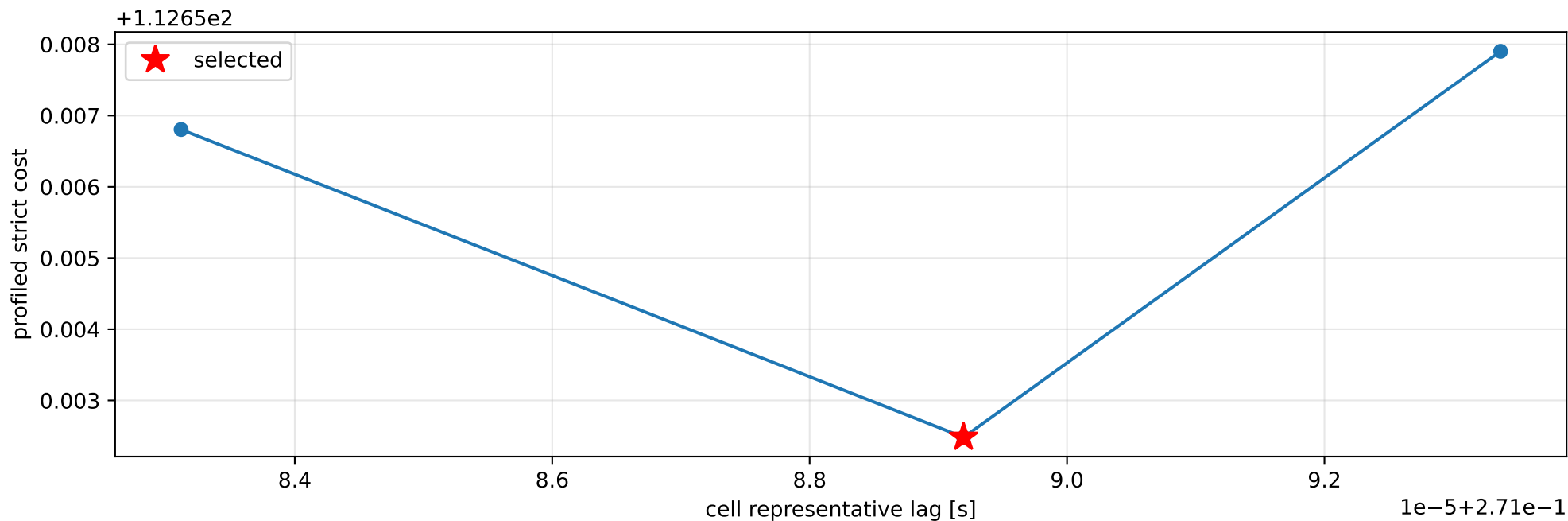
specific force [m/s^2]



kkt_scale_positive: smooth-to-strict continuation



kkt_scale_positive: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

kkt_scale_positive: parameter estimate

Case: kkt_scale_positive

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.52043148

inertia [kg m²]:

```
[[ 0.55443704  0.01372501 -0.00323582]
 [ 0.01372501  0.26168333 -0.08172244]
 [-0.00323582 -0.08172244  0.79125   ]]
```

CoG body [m]: [-0.026435 0.01133929 0.00941535]

force effectiveness: [2.87131352 2.32387838 2.56341465 3.07927631]

scale-free J/m [m²]:

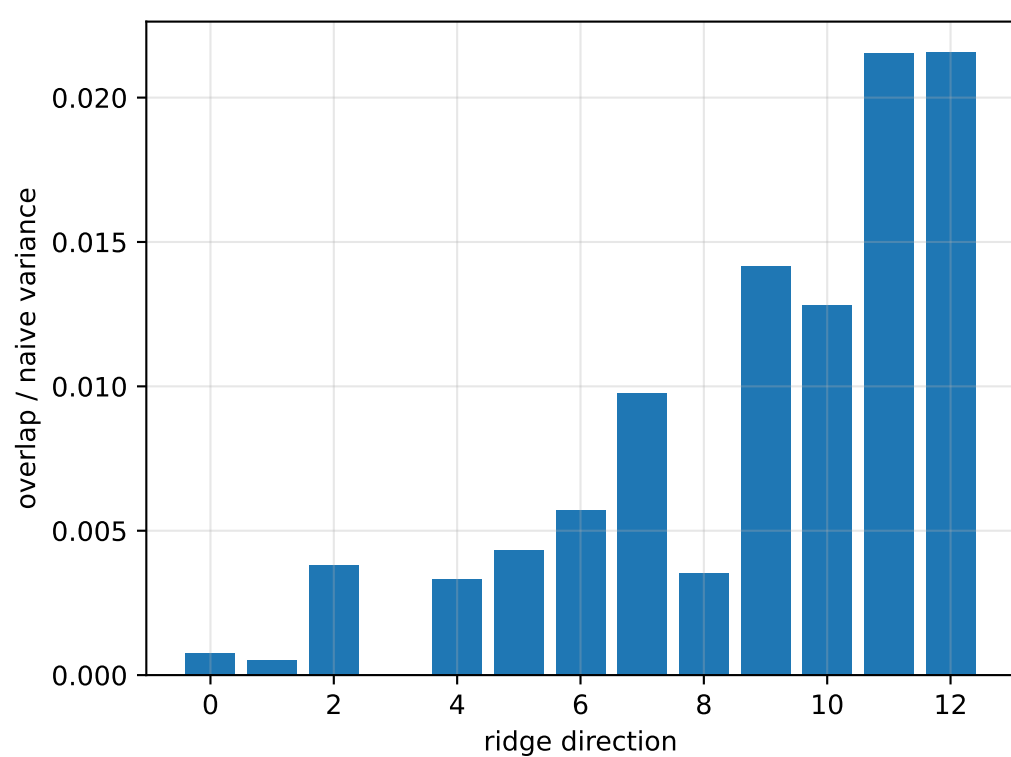
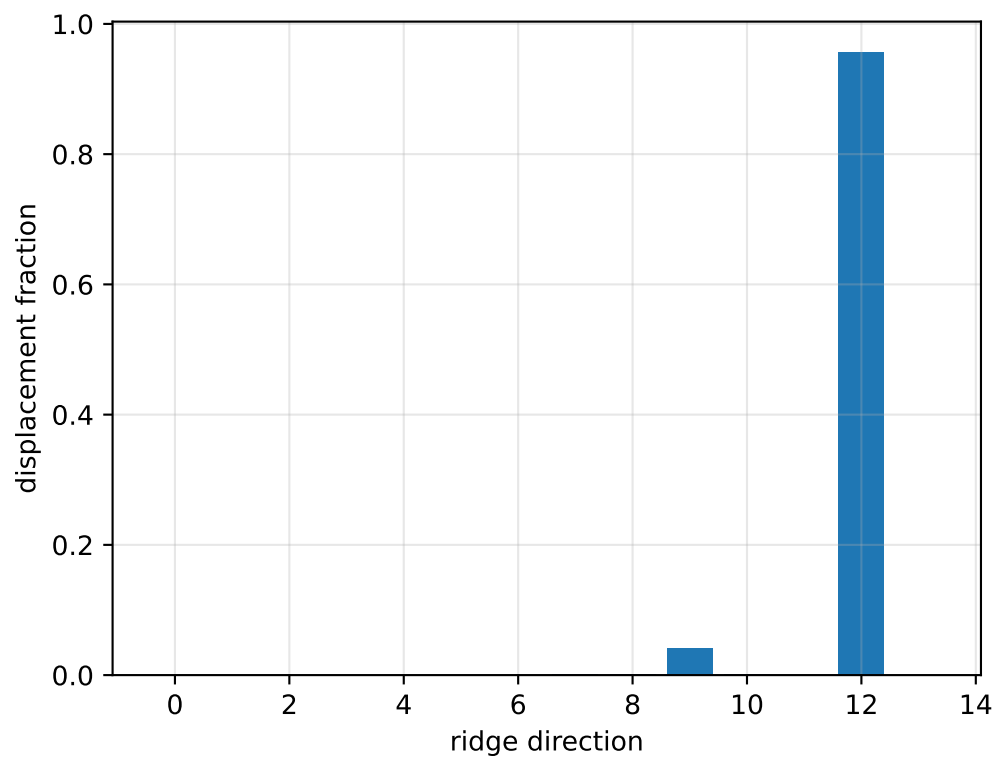
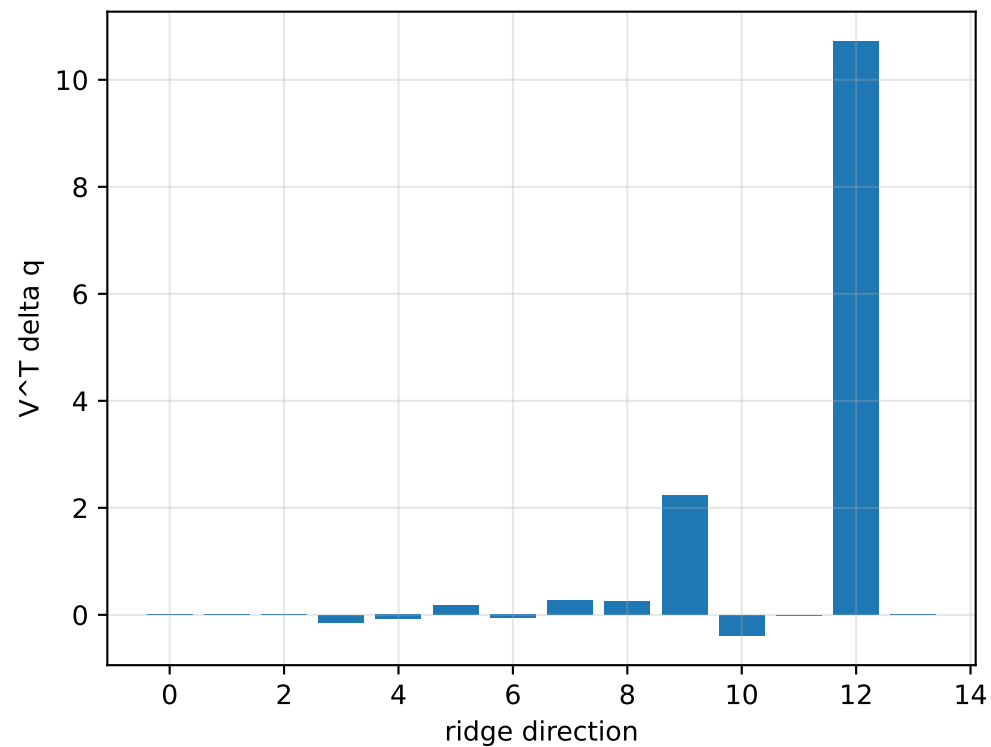
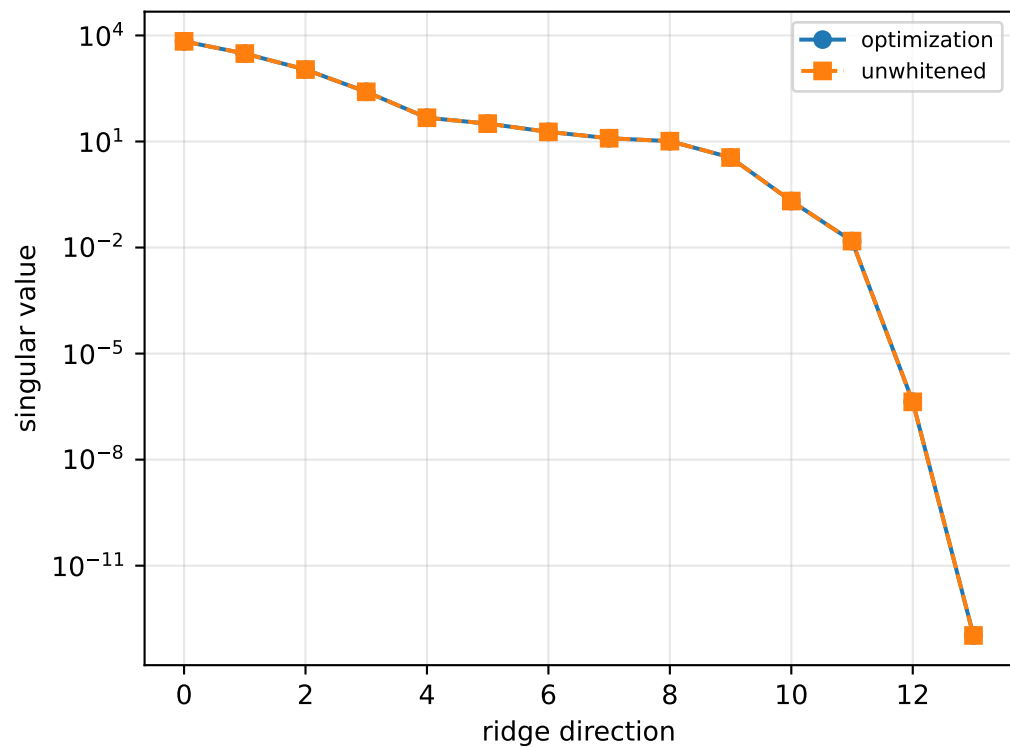
```
[[ 0.0737241  0.00182503 -0.00043027]
 [ 0.00182503  0.03479632 -0.01086672]
 [-0.00043027 -0.01086672  0.10521338]]
```

scale-free f/m: [0.3818017 0.30900865 0.34086005 0.40945474]

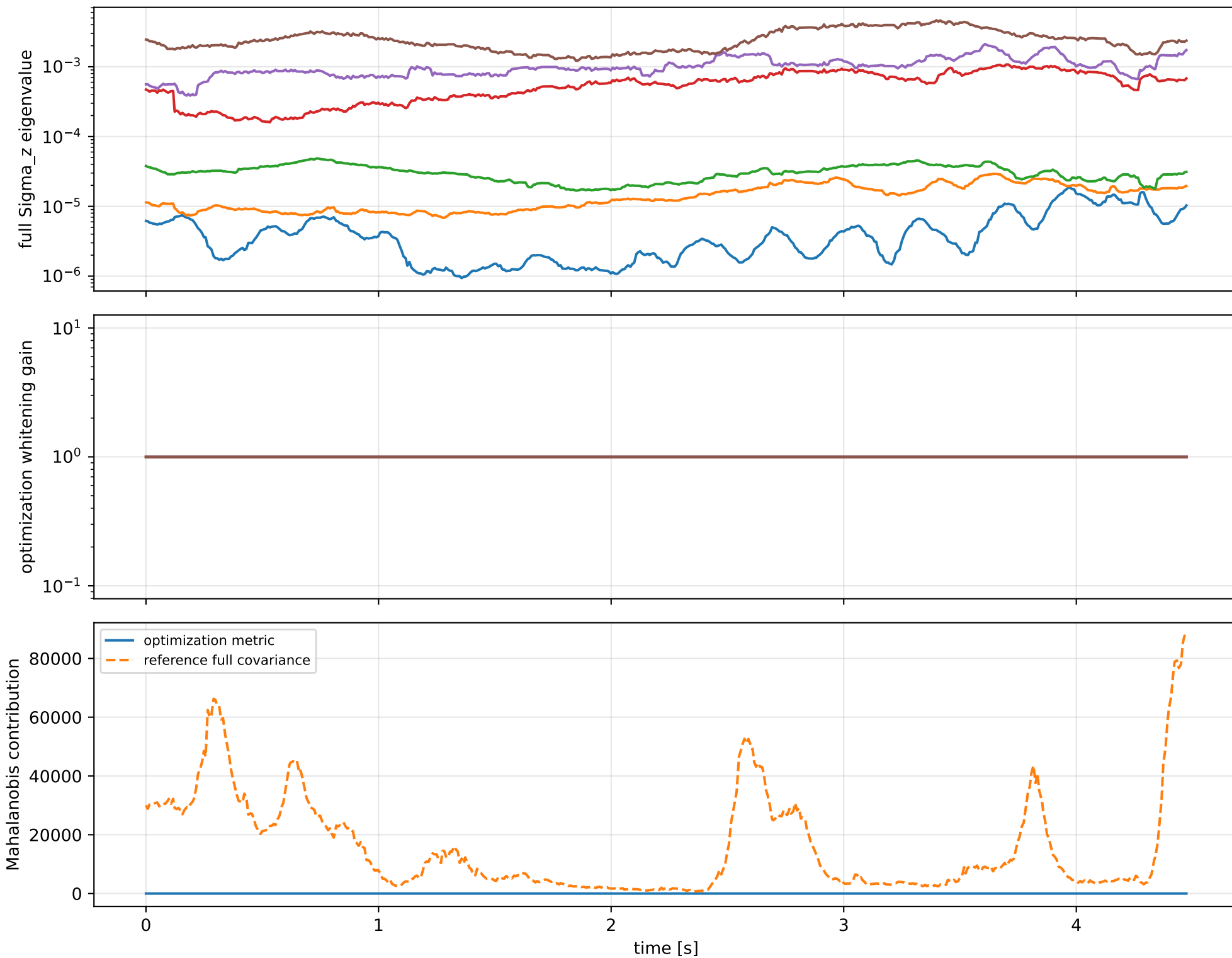
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

kkt_scale_positive: ridge spectrum (rank 13, nullity 1, ||Jv||=1.51e-13)



kkt_scale_positive: optimization vs reference covariance



kkt_scale_positive: wrench / closure (force max 1.98e-14, torque max 4.37e-16)

